

CORE CALCULATIONS and ENERGY RECOVERY (from inductors)

1. INTRODUCTION

Since we are studying circuits that convert electrical power, it makes sense that an extremely important area needed to understand Power Electronics is Power Calculations.

Be careful, it is easy from now on to become confused with all the equations that pop up and lose sight of ‘the big picture’. Try to keep in mind that electrical power is simply the product relationship between voltage and current. The equations are there to ultimately describe how the power (and therefore current and voltage) varies in time.

Also make sure that after this Power Calculations section, you have a good understanding of the current and voltage behaviour of inductors, as these (along with the switches and capacitors) form the backbone of many PE circuits.

2. ENERGY AND POWER

2.1 INSTANTANEOUS POWER

As the name suggests, instantaneous power is just the power at an instant in time. The simple relation is then just,

$$p(t) = v(t) i(t).$$

If $p(t)$ has a positive value (current in same direction as positive voltage polarity) it is an energy **source** (outputs power), if $p(t)$ has a negative value (current direction in opposite direction to positive voltage polarity) it is a **sink** of energy (uses power).

2.2 ENERGY

Energy is simply the time integral of power. Often, the symbol given for energy is W (a link to the mechanical concept of “work”). Do not confuse this with the units abbreviation for power of watts, W.

$$W = \int_{t_1}^{t_2} p(t) dt$$

The units of energy is of course joules, J.

2.3 AVERAGE POWER

Many voltages and currents in PE applications are periodic (but not necessarily sinusoidal). The *average power*, P is simply the power averaged over an entire cycle (of period T).

$$P = \frac{1}{T} \int_{t_0}^{t_0+T} p(t) dt = \frac{W}{T}$$

The average power can also be known as *real power* owing to common AC circuit analysis conventions.

Again, positive P means a source, negative P means a sink.

2.4 INDUCTORS AND CAPACITORS

Ideal inductors and capacitors are solely energy storage circuit elements. This means that on average they do not dissipate power, they simply store energy and subsequently release it.

The instantaneous voltage-current relationships for inductors and capacitors are,

$$v_L(t) = L \frac{d i_L(t)}{d t} \quad \text{and} \quad i_C(t) = C \frac{d v_C(t)}{d t}$$

or

$$i_L(t) = \frac{1}{L} \int_{t_1}^{t_2} v_L(t) dt + i_L(t_1) \quad v_C(t) = \frac{1}{C} \int_{t_1}^{t_2} i_C(t) dt + v_C(t_1)$$

Remember that the instantaneous power is still $p(t) = v(t) i(t)$.

The instantaneous stored energy for inductors and capacitors are,

$$w_L(t) = \frac{1}{2} L i^2(t) \quad \text{and} \quad w_C(t) = \frac{1}{2} C v^2(t)$$

For **periodic** voltages and currents, the special average conditions for inductors and capacitors are summarized below,

$$P_L = 0 \quad P_C = 0$$

$$V_L = 0 \quad I_C = 0$$

2.5 ROOT MEAN SQUARE (RMS) VALUES

RMS effective values are used to simplify power computations when periodic signals are being considered, especially power dissipated by resistances.

In general, the expressions for RMS values are given by,

$$V_{RMS} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt} \quad \text{and} \quad I_{RMS} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt}$$

The following table lists most of the common RMS voltage expressions based on the peak voltage value. Current expressions are identical, just replace V with I .

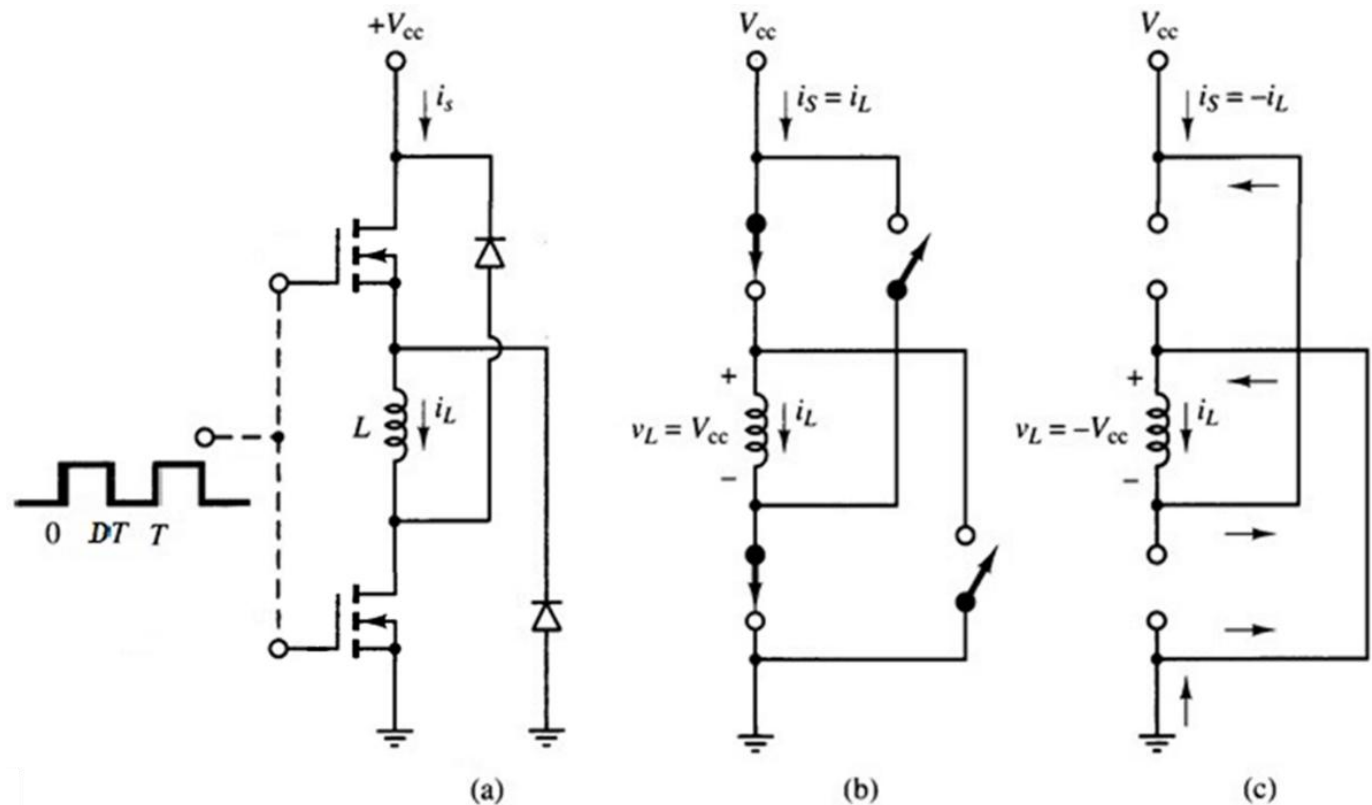
Rectangular pulse (duty cycle D)	$V_{RMS} = V_p \sqrt{D}$
Sinusoids	$V_{RMS} = V_p / \sqrt{2}$
Full-wave rectified sinusoids	$V_{RMS} = V_p / \sqrt{2}$
Triangular wave	$V_{RMS} = V_p / \sqrt{3}$

2.6 ENERGY RECOVERY

The concept of energy recovery is of particular importance in PE. Switching of voltage, current, and energy stored in inductors (and capacitors) provides the functional behaviour needed for PE applications. To efficiently convert power, stored energy needs to be utilised and not simply dissipated as heat in resistive components in the PE circuit.

The circuit shown provides an excellent example of energy recovery from an energised inductor.

Let's look at this circuit in the two separate conditions of when the transistors are **on**, and when they are **off**.



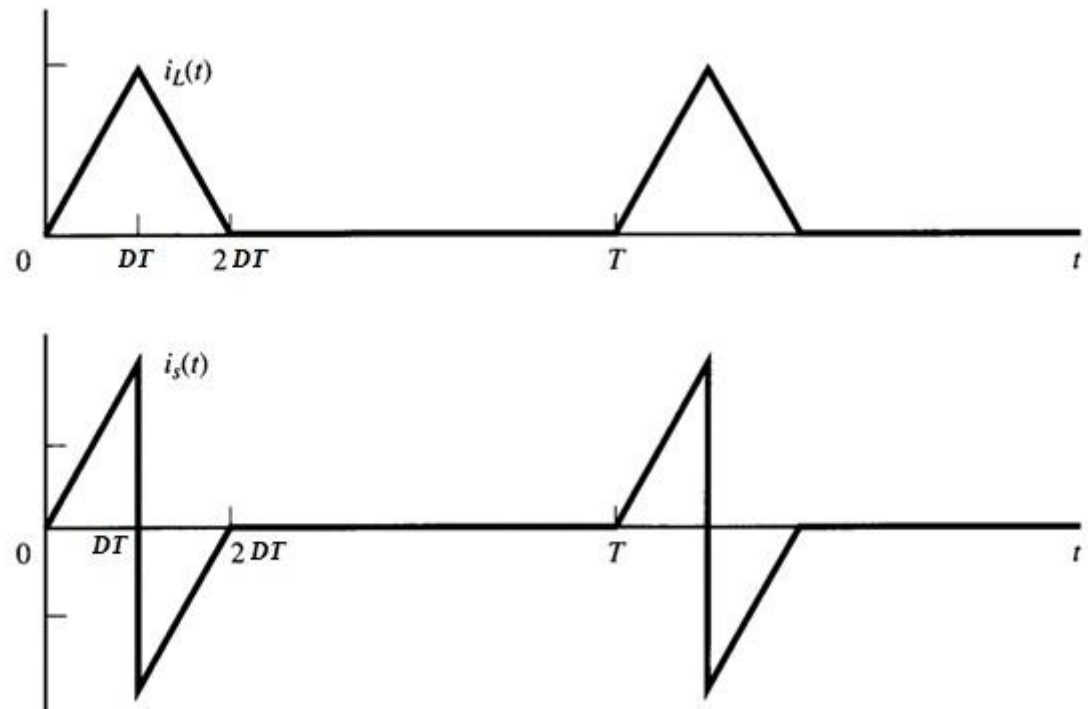
Transistors on: $0 < t < DT$

- Both diodes reverse biased
- Inductor being energised by source current $i_s(t) = i_L(t)$ (sink of energy)
- $v_L(t) = V_{CC}$
- Solving for $i_L(t)$ gives $\frac{V_{CC}t}{L}$

Transistors off: $DT < t < T$

- $v_L(t)$ reversed ($-V_{CC}$), but $i_L(t)$ in same direction – Inductor is now a source of energy
- Both diodes now forward biased
- Solving for $i_L(t)$ gives

- $i_L(t)$ now feeds **into** source so energy is recovered



Point to consider – What happens if an energised inductor (so has current flowing through it such that it is a sink of energy) is suddenly open-circuited?

